

# MASK WHAT MATTERS: CONTROLLABLE TEXT-GUIDED MASKING FOR SELF-SUPERVISED MEDICAL IMAGE ANALYSIS

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## ABSTRACT

The scarcity of annotated data in specialized domains such as medical imaging presents significant challenges to training robust vision models. While self-supervised masked image modeling (MIM) offers a promising solution, existing approaches largely rely on random high-ratio masking, leading to inefficiency and poor semantic alignment. Moreover, region-aware variants typically depend on reconstruction heuristics or supervised signals, limiting their adaptability across tasks and modalities. We propose Mask What Matters, a controllable text-guided masking framework for self-supervised medical image analysis. By leveraging vision-language models for prompt-based region localization, our method flexibly applies differentiated masking to emphasize diagnostically relevant regions while reducing redundancy in background areas. This controllable design enables better semantic alignment, improved representation learning, and stronger cross-task generalizability. We benchmark Mask What Matters against strong MIM and contrastive baselines under identical pretraining and fine-tuning protocols, spanning three distinct imaging modalities (brain MRI, chest CT, and lung X-ray) and three downstream tasks (classification, detection, and instance segmentation). The method wins on every modality and every task: relative to the strongest MIM competitor, SparK, it improves fine-tuning classification accuracy by up to +3.1 percentage points, box average precision (BoxAP) by +1.3, and mask average precision (MaskAP) by +1.1. Crucially, these consistent margins are obtained while masking far less of the image—40% versus the 70% used by the baselines—so the gains come with, rather than at the cost of, greater efficiency. Taken together, these results show that controllable, text-driven masking delivers semantically aligned self-supervised learning that is both more accurate and more efficient, advancing the development of robust vision models for medical image analysis.

## 1 INTRODUCTION

Image classification, segmentation, and object detection are fundamental tasks in computer vision. In recent years, machine learning—especially deep learning—has become the core technology driving advances in this field. However, high-performing deep learning models typically rely on large-scale annotated datasets. In specialized domains such as medical image processing, high-quality annotations are often scarce and expensive to obtain, posing a significant barrier to the widespread adoption of deep learning methods.

Self-supervised learning (SSL) offers a promising solution by using unlabeled data for pretraining, enabling models to learn effective feature representations without manual annotations [Jaiswal et al. (2020); Liu et al. (2021); Krishnan et al. (2022); Huang et al. (2023)]. Among various SSL paradigms, MIM has emerged as a popular approach. By randomly masking parts of an image and requiring the model to reconstruct the original content, MIM encourages the model to capture structural and semantic patterns from visible context. It has demonstrated strong performance across a range of downstream tasks and has been successfully extended to medical image analysis [He et al. (2022); Gupta et al. (2025); Xiao et al. (2023)].

However, prevailing MIM approaches, such as masked autoencoders (MAE) [He et al. \(2022\)](#) and SparK [Tian et al. \(2023\)](#)—typically adopt random masking strategies, which lack alignment with regions of interest in downstream tasks. This task-agnostic masking introduces two concrete limitations that our experiments later quantify: (1) The learned representations may be semantically misaligned with the downstream objectives, reducing transfer performance. (2) To increase the chance of masking informative areas, existing methods often push the masking ratio very high (e.g., 75% in MAE), which increases reconstruction difficulty and demands large-scale data and computational resources. By contrast, natural language is compact and semantically dense, so masked language models learn effectively at far lower ratios [Devlin et al. \(2019\)](#). The reliance on high masking ratios in image-based MIM largely stems from inherent information redundancy and the lack of semantic-aware masking, which makes it difficult for models to focus on truly critical regions. A central empirical question of this paper is therefore whether semantically targeted masking can match or exceed high-ratio random masking while masking substantially less of the image—a question we answer quantitatively in our experiments, where MWM at a 40% ratio surpasses SparK’s best 70% configuration by +2.4 points on chest CT.

This issue is particularly pronounced in medical imaging, and it is precisely where a semantically guided strategy should pay off most. Medical images exhibit high semantic sparsity: diagnostically relevant information is often confined to small localized regions (e.g., lesions or organs), while a large fraction of the image is semantically redundant background. Medical imaging also spans diverse modalities—MRI, CT, and X-ray—with substantial variation in the shape, size, and spatial distribution of task-relevant areas across tasks. These properties raise the bar for both the efficiency and the generalizability of a masking strategy, and they make cross-modality consistency—rather than a single headline number—the decisive measure of success. We therefore evaluate every claim across all three modalities and report the margin against the strongest competing baseline in each case.

To address these challenges, we introduce a controllable, text-guided masking framework—hereafter referred to as **Mask-What-Matters (MWM)**. Given a user-specified, open-vocabulary description or phrase, a pretrained vision–language model provides zero-shot localization cues that we convert into robust regions of interest (ROIs). MWM then applies region-specific, prompt-conditioned masking—assigning higher ratios to semantically important areas (e.g., lesions/organs) and lower ratios to background—thereby injecting downstream semantics directly into the pretraining signal without per-image reports or labels.

To summarize, our main contributions are:

- The first controllable text-guided masking framework for medical imaging. MWM integrates vision–language models (BiomedCLIP) with segmentation refinement (SAM) to localize task-relevant regions from open-vocabulary prompts and apply differentiated ROI vs. background masking, overcoming the semantic misalignment of random masking.
- An annotation-free and backbone-agnostic design. MWM requires no per-image reports or labels, and can be seamlessly integrated into both ViT- and ConvNet-based MIM pipelines under a unified protocol.
- Consistent, measurable gains across every modality and task tested. Under matched pretraining and fine-tuning protocols, MWM outperforms the strongest MIM baselines (e.g., SparK, AnatoMask) on all three datasets (brain MRI, chest CT, lung X-ray) and all three tasks: it leads fine-tuning classification on every modality (up to +3.1 points over SparK), wins linear probing on every modality, and improves detection (+1.3 BoxAP) and segmentation over SparK—while masking only 40% of the image against the baselines’ 70%. This uniform win pattern, rather than an isolated best case, is the central evidence that text-driven generative pretraining is effective in medical imaging.

## 2 RELATED WORK

### 2.1 MASKED IMAGE MODELING

Masked modeling originated in the natural language processing domain with the bidirectional encoder representations from transformers (BERT) model [Devlin et al. \(2019\)](#), which learns contextual representations by masking tokens and predicting them during pretraining. Inspired by this idea, [He](#)

et al. (2022) introduced MIM to computer vision, where portions of an image are masked during pre-training and the model is tasked with reconstructing the missing regions. This paradigm has demonstrated strong generalization in both the natural and medical image domains (Gupta et al. 2025; Liu et al. 2021; Huang et al. 2023). The representative methods against which we later benchmark are MAE (He et al. 2022), MedMAE (Gupta et al. 2025; Xiao et al. 2023), and SparK (Tian et al. 2023). Among them, MAE uses high-ratio random masking to push ViTs toward high-level semantic representations; MedMAE extends this idea to medical images with domain-specific adaptations; and SparK adapts the strategy to convolutional backbones, exploiting their multi-scale features to strengthen pretraining. Notably, all three share the same design choice—random, high-ratio masking—that our experiments identify as the key bottleneck, with SparK serving as the strongest MIM baseline and the primary point of comparison throughout.

Despite these advances, conventional MIM approaches typically adopt random or structure-agnostic masking strategies that ignore the semantic requirements of downstream tasks. Such non-selective masking can lead the model to overemphasize irrelevant regions during pretraining, reducing transfer effectiveness—an effect we later quantify, where random masking (the *No Prompt* setting in Table 5) trails text-guided masking on all three modalities. To close this gap, recent work has explored region-aware masking strategies aimed at improving semantic alignment. Two main directions have emerged, and both leave measurable room for improvement:

One line of work introduces loss-driven adaptive masking, where methods such as hard patches mining (HPM) (Wang et al. (2023a)) and its extension AnatoMask (Li et al. (2024)) dynamically analyze reconstruction errors across regions and assign higher masking probabilities to areas with larger losses, aiming to prioritize semantically valuable structures. However, this approach can mislead the model toward hard-to-reconstruct but task-irrelevant regions. For example, in brain magnetic resonance imaging (MRI), complex structures like the spinal cord or eyeballs may attract unnecessary attention despite being unrelated to downstream tasks like tumor detection.

Another line of work incorporates external perceptual modules to guide masking. For example, FocusMAE (Basu et al. (2024)) leverages pretrained Region Proposal Networks (RPNs) to identify high-information regions for selective masking. While effective in certain tasks, this strategy requires supervised training of RPNs on domain-specific annotated data, limiting its cross-task generalization.

These limitations highlight the need for a more flexible, semantically aligned and generalizable masking mechanism. Natural language, as a rich and interpretable source of supervision, presents a promising direction for guiding masked modeling in a task-aware yet annotation-free manner.

## 2.2 VISION-LANGUAGE PRE-TRAINED MODELS

Benefiting from the rapid development of vision-language alignment models in recent years, it has become increasingly feasible to accurately perceive and identify task-relevant regions under zero-shot and open-world settings (Radford et al. (2021)). Classic approaches such as the Contrastive Language-Image Pretraining (CLIP) family achieve a unified embedding space for images and texts through large-scale joint pretraining on image-text pairs, and have been widely adopted in zero-shot and open-set visual tasks. In the medical imaging domain, extensions like BiomedCLIP (Zhang et al. (2023)) further enhance cross-modal understanding, achieving robust performance across diverse downstream applications. Although vision-language pretrained models have been widely adopted in various supervised learning tasks, most existing studies still confine them to downstream applications such as object detection and semantic segmentation (Zhong et al. 2022; Aleem et al. 2024; Koleilat et al. 2024), leaving open the question of whether their open-vocabulary and zero-shot capabilities can benefit masked image modeling. Our localization results (Table 1) answer this directly, showing that a frozen BiomedCLIP-driven pipeline recovers task-relevant regions at around 0.82–0.94 occlusion recall across modalities, purely from text.

To summarize, although recent region-aware MIM approaches—such as dynamic masking based on reconstruction loss or masking guided by external perception modules—have improved semantic focus to some extent, they still suffer from several limitations: reliance on additional supervision, misalignment between masked regions and task semantics, and lack of flexibility. Meanwhile, despite the strong performance of vision-language models in downstream tasks, little effort has been made to incorporate natural language prompts into the MIM pretraining phase for task-aware masking

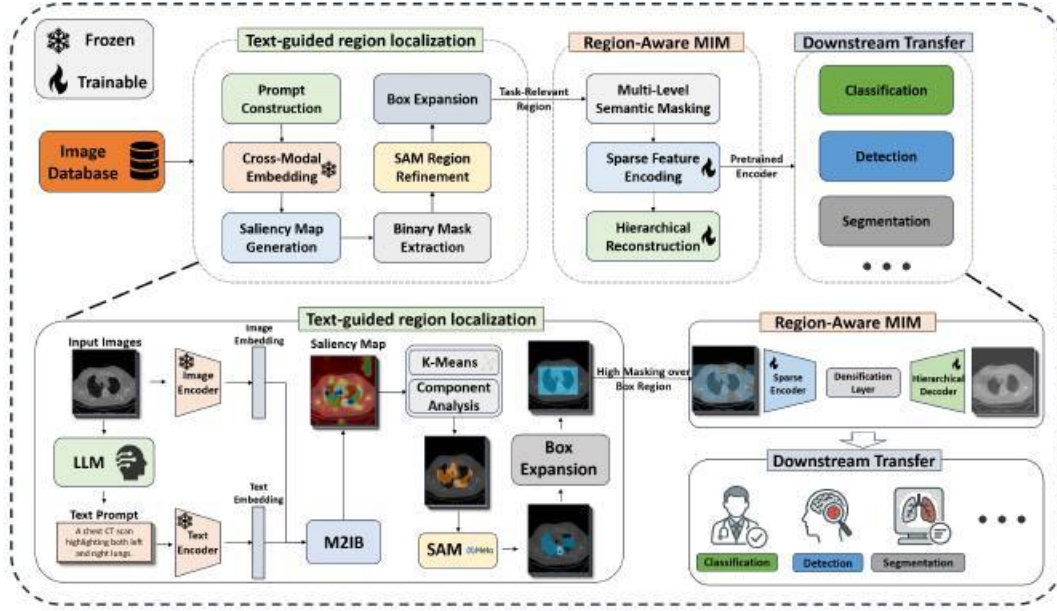


Figure 1: Overview of the MWM framework. The top panel shows the three-stage pipeline: (1) text-guided region localization, (2) region-aware masked image modeling, and (3) downstream transfer, where the learned encoder is evaluated on classification, detection, and segmentation. The bottom panel zooms in to illustrate how prompts guide region localization and how the resulting ROI drives differentiated (high-ratio foreground vs. light background) masking—the mechanism responsible for the accuracy and efficiency gains reported in Tables 2-4

guidance. Consequently, these observations point to a promising direction—incorporating vision-language semantic guidance to enable efficient and generalizable masked image modeling.

### 3 METHODOLOGY

#### 3.1 OVERALL FRAMEWORK

As illustrated in Figure 1, **Mask What Matters (MWM)** comprises two core stages: (1) text-guided region localization and (2) region-aware masked image modeling. The first stage leverages a frozen vision-language model to identify task-relevant regions from an open-vocabulary prompt, while the second stage applies differentiated masking ratios across those regions to steer reconstruction. Together these two stages are what let MWM concentrate the learning signal on diagnostically important pixels; our ablations isolate the contribution of each stage—prompt design and SAM refinement in Table 5, and masking ratio in Figure 4—and show that both translate into measurable downstream gains.

#### 3.2 TEXT-GUIDED REGION LOCALIZATION

MWM localizes task-relevant regions through a multi-stage pipeline. It begins with LLM-based prompt generation and employs BiomedCLIP to extract cross-modal embeddings. A saliency map is then produced via the Multi-Modal Information Bottleneck (M2IB), binarized with K-Means, and refined using the Segment Anything Model (SAM). Finally, the refined mask is converted into a bounding box with a controllable expansion margin, which serves as a robust spatial guide for the subsequent masked modeling stage.

**Prompt Generation.** In medical image analysis, the design of textual prompts strongly influences how well regions are localized. To strengthen semantic guidance, large language models (LLMs) are queried with structured prompts such as “Describe the typical visual characteristics of a [category] in a [modality] image.” As an alternative, simple category phrases are also used as inputs. These

two prompt styles are compared head-to-head in the experimental section—both at the localization level (Table 1) and in downstream accuracy (Table 5)—and the comparison shows that neither style dominates universally: descriptive sentences win on brain MRI and chest CT, while concise phrases are stronger on lung X-ray.

**Cross-Modal Embedding.** Once the prompt is generated, a BiomedCLIP model [Zhang et al., 2023] encodes the medical image and its prompt into a shared semantic space—the frozen, annotation-free backbone whose zero-shot localization quality we quantify in Table 1. Given an image  $I$  and a prompt  $T$ , the embeddings are computed as:

$$Z_{\text{img}} = \Phi_{\text{img}}(I), \quad Z_{\text{text}} = \Phi_{\text{text}}(T) \quad (1)$$

where  $\Phi_{\text{img}}$  and  $\Phi_{\text{text}}$  denote the vision and language encoders, respectively. These cross-modal embeddings provide the semantic foundation for saliency estimation in subsequent stages.

**Saliency Map Generation.** A cross-modal saliency estimation module based on the Multi-modal Information Bottleneck (M2IB) [Wang et al., 2023b] is exploited to localize task-relevant visual regions by aligning image and text embeddings while suppressing modality-specific redundancy. Formally, the saliency map  $\lambda_S \in [0, 1]^{H \times W}$  assigns importance weights to each spatial location based on its semantic relevance to the prompt. M2IB achieves this by optimizing the following objective:

$$\mathcal{L}_{\text{M2IB}} = \text{MI}(Z_{\text{img}}, Z_{\text{text}}) - \gamma \cdot \text{MI}(Z_{\text{img}}, I) \quad (2)$$

where  $Z_{\text{img}}$  and  $Z_{\text{text}}$  are image and text embeddings encoded by BiomedCLIP respectively, and  $\gamma$  controls the trade-off between preserving cross-modal relevance and filtering out task-irrelevant visual information.

**Binary Mask Extraction.** Given the saliency map, we binarize it with unsupervised K-Means clustering [Lloyd (1982)] to localize the prompt-guided region and produce a preliminary foreground mask. The map is clustered into two pixel groups, and the higher-saliency cluster is taken as foreground. To suppress false activations, connected component analysis then retains only the largest foreground regions. This step filters out noisy edge areas and concentrates attention on core semantic structures—precision that the subsequent SAM stage measurably sharpens (e.g., localization precision rises from 0.22 to 0.30 on brain MRI; Table 1).

**Region Refinement.** To enhance spatial accuracy, SAM is incorporated as a refinement module [Kirillov et al., 2023]. For each selected connected region  $c_i \in C^*$ , its minimal enclosing bounding box is computed to serve as the visual prompt for SAM.

Given the original image  $I$  and the set of bounding boxes, SAM predicts a refined segmentation mask:

$$M_{\text{SAM}} = \text{SAM}(I, \text{Box}(c_i)) \quad (3)$$

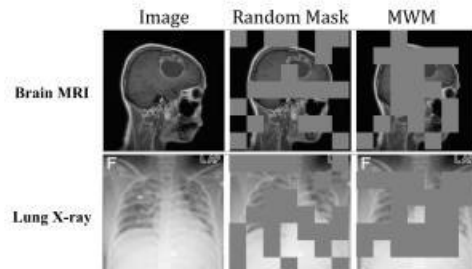
The resulting mask  $M_{\text{SAM}}$  provides more faithful spatial structure.

**Box-Based Region Expansion.** The SAM mask from the previous stage still carries boundary noise that can mislead the model’s focus during representation learning. To make region guidance more robust, the refined SAM mask is converted into a bounding box with a controllable expansion margin, which becomes the final region of interest (ROI) for downstream masked modeling. This box-level ROI is what preserves the high occlusion recall reported in Table 1 (around 0.82–0.94) while keeping precision high enough to drive the downstream accuracy gains.

### 3.3 REGION-AWARE MASKED IMAGE MODELING

Given the region of interest generated by the localization module in Section 3.2, MWM performs masked image modeling through a three-stage pipeline: (1) a multi-level masking strategy that applies differentiated masking ratios based on region importance, (2) a sparse encoder that processes only the unmasked patches, and (3) a hierarchical decoder that reconstructs the full image from sparse multi-scale features.

**Multi-Level Semantic Masking.** MWM adopts a multi-level masking strategy built on the semantic regions from the text-driven localization stage. Patches inside prompt-highlighted regions—those



associated with tumors or organs—are masked at a high ratio so the model must reconstruct task-relevant semantic features, whereas background regions are masked lightly or left visible to cut redundant computation. This asymmetry is the source of MWM’s efficiency advantage: it concentrates the reconstruction objective where signal is dense, which is why an overall 40% ratio suffices to beat SparK’s best 70% setting by +2.4 points on chest CT (Figure 4). Figure 2 contrasts the resulting masking pattern with conventional uniform random masking.

**Sparse Feature Encoding.** Visible patches are collected into a sparse image and encoded following the SparK design Tian et al. (2023):

$$\begin{aligned} X_{\text{sparse}} &= \{x_i \mid m_i = 1\}, \\ F &= \Phi_{\text{sparse}}(X_{\text{sparse}}) \end{aligned} \quad (4)$$

where  $x_i$  is the  $i$ -th patch,  $m_i \in \{0, 1\}$  indicates visibility,  $\Phi_{\text{sparse}}$  is the encoder, and  $F = \{F_1, F_2, F_3, F_4\}$  are multi-scale features. While we instantiate MWM with convolutional backbones (ResNet-50), the same interface applies to ViTs via token indexing; this backbone-agnostic design is what allows the like-for-like comparison against both convolutional (SparK) and transformer (MAE) baselines at matched model scale reported in Table 2

**Hierarchical Reconstruction.** A lightweight UNet-style decoder reconstructs the full image in a top-down manner. At each scale  $l$ , masked locations are filled with a learned embedding  $M_l$ :

$$F'_l(i) = \begin{cases} F_l(i), & \text{if patch } i \text{ is visible,} \\ M_l, & \text{if patch } i \text{ is masked.} \end{cases} \quad (5)$$

The densified map  $F'_l$  is projected by a scale-specific layer  $\phi_l$  to obtain  $D_l$ , then upsampled and fused with features at the next lower level:

$$D_{l-1} = B_{l-1}(D_l) + \phi_{l-1}(F'_{l-1}). \quad (6)$$

Finally, the model is trained with mean-squared error on masked patches:

$$\mathcal{L}_{\text{recon}} = \frac{1}{|\mathcal{I}_{\text{mask}}|} \sum_{i \in \mathcal{I}_{\text{mask}}} \|\hat{x}_i - x_i\|^2, \quad (7)$$

where  $\mathcal{I}_{\text{mask}}$  is the set of masked patch indices,  $x_i$  the original patch, and  $\hat{x}_i$  its reconstruction.

## 4 EXPERIMENTS

### 4.1 DATASETS

**Self-Supervised Pretraining Datasets.** To test generalizability rather than a single-modality result, pretraining spans three distinct imaging modalities. (1) *Brain MRI*: approximately 17,000 images from Brain Tumor MRI Cheng et al. (2015), BRISC Fateh et al. (2025), and BraTS 2018 Menze et al. (2014), covering diverse tumor types and anatomical structures. (2) *Lung X-ray*: around 39,000 chest radiographs from COVID-QU-Ex Rahman et al. (2021); Chowdhury et al. (2020), spanning COVID-19, non-COVID pneumonia, and normal cases. (3) *Chest CT*: 12,000 slices from a lung disease dataset Konya (2020), with segmentation masks of fibrotic lungs from 107 patients.

**Downstream Fine-Tuning Datasets.** Downstream evaluation uses one dataset per modality, again spanning all three modalities so that every reported comparison is repeated three times rather than on a single benchmark. (1) *Brain Tumor MRI* Feltrin (2023): 4,479 MRI images across 44 tumor types, including astrocytoma, glioblastoma, meningioma, and ependymoma. (2) *Pediatric Lung X-ray* Kermany et al. (2018): 5,863 pediatric chest X-rays labeled normal or pneumonia. (3) *Chest CT Cancer* Hany (2020): CT images from four classes—adenocarcinoma, large cell carcinoma, squamous cell carcinoma, and normal tissue.

| Method               | Brain MRI   |             | Chest CT    |             | Lung X-ray  |             |
|----------------------|-------------|-------------|-------------|-------------|-------------|-------------|
|                      | Prec.       | Rec.        | Prec.       | Rec.        | Prec.       | Rec.        |
| <i>Prompt Type</i>   |             |             |             |             |             |             |
| Phrase               | 0.22        | 0.86        | 0.21        | <b>0.99</b> | 0.42        | <b>0.82</b> |
| Descrip. Sent.       | <b>0.30</b> | <b>0.94</b> | <b>0.37</b> | 0.88        | <b>0.44</b> | 0.66        |
| <i>Impact of SAM</i> |             |             |             |             |             |             |
| Coarse Ext.          | 0.22        | <b>0.95</b> | 0.36        | 0.87        | 0.37        | <b>0.84</b> |
| + SAM Ext.           | <b>0.30</b> | 0.94        | <b>0.37</b> | <b>0.88</b> | <b>0.42</b> | 0.82        |

Table 1: Text-guided localization performance (occlusion Precision/Recall; higher is better) across the three modalities, broken out by prompt type and by refinement strategy. Best value in each block/column is in **bold**. *Coarse Ext.* denotes bounding-box extension of the coarse region; + *SAM Ext.* adds SAM refinement. Two patterns stand out: descriptive sentences raise precision on brain MRI (0.22→0.30) and chest CT (0.21→0.37), and adding SAM raises precision while holding recall essentially fixed (e.g., brain MRI precision 0.22→0.30 at recall ≥0.94).

## 4.2 EXPERIMENTAL SETUP

To keep every comparison fair, all self-supervised methods use backbones of matched model scale—convolution-based methods adopt ResNet-50 [He et al. \(2016\)](#) and transformer-based methods ViT-S [Dosovitskiy et al. \(2020\)](#)—with a common input resolution of 224×224 and identical ImageNet-pretrained initialization. Pretraining uses an initial learning rate of  $2 \times 10^{-4}$ , and downstream fine-tuning retains the same encoder and input resolution. All experiments run on NVIDIA Tesla A100 cards (80 GB VRAM) under Python 3.9, Ubuntu 22.04.3, and PyTorch 1.10.0. Unless otherwise specified, non-target variables (e.g., prompt type or SAM usage) are held at their empirically optimal settings so that each reported difference is attributable to the single factor under study. This controlled protocol means the margins reported below—e.g., MWM over SparK and AnatoMask—reflect the masking strategy itself rather than differences in scale, initialization, or tuning.

## 4.3 EVALUATION OF TEXT-GUIDED LOCALIZATION

We first quantify the text-guided localization module across all three modalities—brain MRI, lung X-ray, and chest CT—and isolate the effect of prompt type and SAM refinement. Predicted bounding boxes are compared against expert-annotated segmentation masks using occlusion precision ( $|R_p \cap R_{gt}|/|R_p|$ ) and occlusion recall ( $|R_p \cap R_{gt}|/|R_{gt}|$ ), where  $R_p$  is the predicted region and  $R_{gt}$  the ground-truth annotation.

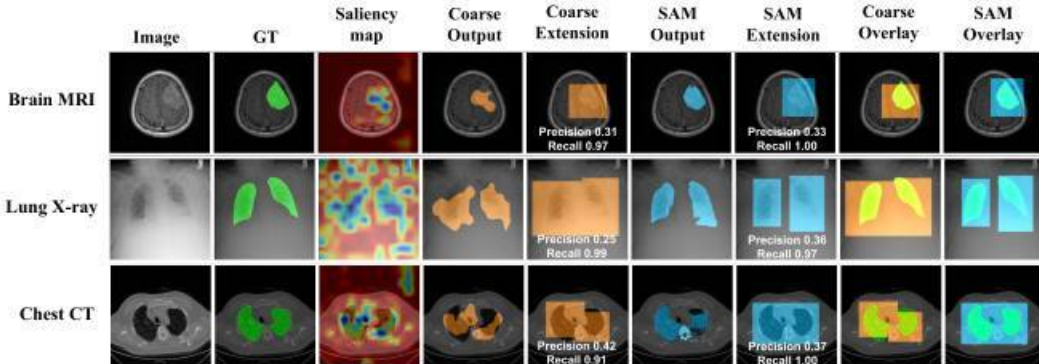


Figure 3: Representative results of text-guided region localization, illustrating the qualitative accuracy behind the quantitative occlusion precision/recall reported in Table 1. Across brain MRI, chest CT, and lung X-ray, predicted regions closely track the diagnostically relevant structures.

The headline finding in Table 1 is consistency: with effective prompts the module reaches high occlusion recall on every dataset (around 0.82, up to 0.94 on brain MRI) while keeping precision reasonable. In practical terms, this means the box almost always contains the diagnostically relevant structure—the property that matters for guiding masking—and it holds across three anatomically very different modalities using text alone (see Figure 3 for qualitative examples).

**Effect of Prompt Type on Region Localization.** Table 1 shows that prompt design has a clear, modality-dependent effect. Descriptive sentence prompts are the stronger choice on brain MRI (precision 0.30 vs. 0.22) and chest CT (precision 0.37 vs. 0.21), where richer semantics sharpen localization. On lung radiographs the ranking flips: concise phrases recover far more of the target region (recall 0.82 vs. 0.66), plausibly because the lungs’ broad anatomical coverage makes general terms sufficient. The practical takeaway is that prompt style should be matched to the modality, and either style already localizes the region well.

**Effect of SAM Refinement on Region Localization.** Applying SAM to the coarse masks yields a clean precision-for-free result: precision rises while recall is essentially preserved. On brain MRI, precision climbs from 0.22 to 0.30 (a +0.08 gain) with recall staying above 0.94, and chest CT and lung X-ray precision likewise improve (to 0.37 and 0.42). Because recall does not degrade, SAM tightens the predicted region without discarding target content—exactly the behavior needed for reliable downstream masking.

#### 4.4 COMPARISON WITH PREVIOUS SSL METHODS

We now benchmark MWM’s representation quality against a broad panel of self-supervised baselines drawn from both paradigms—masked image modeling (MAE, AnatoMask, SparK) and contrastive learning (MoCoV2, BYOL, SimCLR). All methods are pretrained and evaluated under identical datasets and training settings, so the comparison isolates the learning objective. Across the tables that follow, SparK is the strongest prior MIM method and serves as our primary reference point; we report MWM’s margin over it, and over the best-performing baseline overall, for every modality and task.

**Image Classification.** We report classification under the two standard protocols: (1) *full fine-tuning*, which updates the entire model, and (2) *linear probing*, which freezes the encoder and trains only a linear head—a stricter probe of representation quality.

- **Full fine-tuning:** Table 2 compares MWM against six representative baselines spanning both paradigms—MIM methods MAE [He et al., (2022)], SparK [Tian et al., (2023)], and AnatoMask [Li et al., (2024)], and contrastive methods SimCLR [Chen et al., (2020a)], MoCoV2 [Chen et al., (2020b)], and BYOL [Grill et al., (2020)]. MWM ranks first on all three modalities—96.8% (brain MRI), 97.5% (chest CT), and 96.0% (lung X-ray)—so no baseline matches it on any dataset. The margin is largest on chest CT, where MWM’s 97.5% exceeds the strongest MIM baseline SparK by +3.1 points and AnatoMask by +1.7. This is exactly the modality where diagnostic signal (e.g., lung lobes) is most spatially sparse, which is where emphasizing semantically critical regions while skipping redundant background should help most—and the numbers bear that out.

| Method           | Type       | Brain MRI   | Chest CT    | Lung X-ray  |
|------------------|------------|-------------|-------------|-------------|
| MoCoV2           | CL         | 95.3        | 91.7        | 93.1        |
| BYOL             | CL         | 91.3        | 93.7        | 93.9        |
| SimCLR           | CL         | 94.5        | 93.1        | 94.1        |
| MAE              | MIM        | 95.4        | 91.7        | 94.9        |
| AnatoMask        | MIM        | 96.5        | 95.8        | 95.2        |
| SparK            | MIM        | 96.2        | 94.4        | 94.7        |
| <b>MWM(Ours)</b> | <b>MIM</b> | <b>96.8</b> | <b>97.5</b> | <b>96.0</b> |

Table 2: Fine-tuning classification accuracy (%) on the three downstream datasets (higher is better; best per column in **bold**); MIM denotes Masked Image Modeling and CL Contrastive Learning. MWM ranks first on every modality, with the largest margin on chest CT (97.5%, i.e., +3.1 over SparK and +1.7 over AnatoMask).

| Method           | Brain MRI   | Chest CT    | Lung X-ray  |
|------------------|-------------|-------------|-------------|
| AnatoMask        | 70.2        | 63.9        | 85.3        |
| SparK            | 69.8        | 63.5        | 81.6        |
| <b>MWM(Ours)</b> | <b>71.3</b> | <b>67.0</b> | <b>88.5</b> |

Table 3: Linear probing accuracy (%) with a frozen encoder (higher is better; best per column in **bold**). MWM leads on all three modalities, improving over SparK by +1.5 (brain MRI), +3.5 (chest CT), and +6.9 (lung X-ray)—larger margins than under full fine-tuning, indicating the encoder itself learns more transferable features.

| Method            | Detection   |                 | Segmentation |                  |
|-------------------|-------------|-----------------|--------------|------------------|
|                   | $AP^{box}$  | $AP_{75}^{box}$ | $AP^{mask}$  | $AP_{75}^{mask}$ |
| AnatoMask         | 45.2        | 53.1            | <b>46.2</b>  | 54.1             |
| SparK             | 45.6        | 53.2            | 44.8         | 52.0             |
| <b>MWM (Ours)</b> | <b>46.9</b> | <b>53.5</b>     | 45.9         | <b>54.1</b>      |

Table 4: Detection ( $AP^{box}$ ,  $AP_{75}^{box}$ ) and instance segmentation ( $AP^{mask}$ ,  $AP_{75}^{mask}$ ) on the BR35H dataset (%; higher is better; best per column in **bold**). MWM leads on three of four metrics and ties for the best  $AP_{75}^{mask}$  (54.1); relative to SparK it gains +1.3  $AP^{box}$  and +1.1  $AP^{mask}$ , confirming that the classification gains carry over to localization- and pixel-level tasks.

- **Linear probing:** The frozen-encoder setting in Table 3 tells a sharper version of the same story. Against the two strongest MIM baselines, SparK and AnatoMask, MWM wins on every modality, and its lead over SparK actually widens relative to fine-tuning: +1.5 on brain MRI (71.3 vs. 69.8), +3.5 on chest CT (67.0 vs. 63.5), and +6.9 on lung X-ray (88.5 vs. 81.6). Because linear probing rewards features that are already discriminative before adaptation, these larger margins indicate that the gains come from a genuinely better encoder rather than from fine-tuning.

**Object Detection and Instance Segmentation.** To test whether the representation gains extend beyond classification, we evaluate brain tumor detection and instance segmentation on the BR35H dataset Hamada (2025). Table 4 shows MWM taking the top  $AP^{box}$  (46.9%, i.e., +1.3 over SparK) and matching the best  $AP_{75}^{box}$  (53.5) and  $AP^{mask}$  (54.1). On  $AP^{mask}$  it reaches 45.9%—+1.1 over SparK and within 0.3 of AnatoMask’s best. In short, MWM is first or tied-first on three of the four detection/segmentation metrics, so the advantage established in classification transfers cleanly to object-level localization and pixel-wise segmentation.

Read together, the classification, detection, and segmentation results form a single consistent pattern: text-driven masking during pretraining yields more informative and transferable representations, and the improvement over the strongest baselines holds across every task and modality rather than in any one favorable setting.

## 4.5 ABLATION STUDY

To attribute MWM’s gains to specific design choices, we ablate three factors one at a time—prompt design, SAM-based region refinement, and masking ratio—and measure each factor’s effect on downstream classification.

### 4.5.1 EFFECT OF PROMPT DESIGN AND REGION REFINEMENT

Table 5 makes two effects explicit. First, prompt granularity matters and tracks localization quality: descriptive sentences beat short phrases on brain MRI (96.8 vs. 96.4) and chest CT (97.5 vs. 96.5), while phrases win on lung X-ray (96.0 vs. 95.2)—the same modality-dependent ranking seen in the localization results of Section 4.3, confirming that better region identification yields better downstream accuracy. Second, and most importantly, *every* text-guided variant beats random masking (*No Prompt*) on *every* modality, with the gap reaching +3.1 points on chest CT (97.5 vs. 94.4). This directly quantifies the value of semantic alignment between pretraining and downstream tasks. The SAM ablation adds a third, orthogonal gain: enabling SAM improves accuracy on all three modali-

| Group         | Setting   | Brain MRI   | Chest CT    | Lung X-ray  |
|---------------|-----------|-------------|-------------|-------------|
| Prompt Type   | No Prompt | 96.2        | 94.4        | 94.7        |
|               | Phrase    | 96.4        | 96.5        | <b>96.0</b> |
|               | Sentence  | <b>96.8</b> | <b>97.5</b> | 95.2        |
| Impact of SAM | w/o SAM   | 96.4        | 96.8        | 95.8        |
|               | w/ SAM    | <b>96.8</b> | <b>97.5</b> | <b>96.0</b> |

Table 5: Ablation of prompt type and region refinement on downstream classification accuracy (%) across the three modalities (best per column in **bold**). *No Prompt* is random masking without text guidance; *w/* and *w/o* denote with and without SAM. Every text-guided variant beats *No Prompt* on every modality (e.g., chest CT 94.4→97.5), and adding SAM improves all three modalities (chest CT 96.8→97.5), isolating the contribution of each component.

ties (e.g., chest CT 96.8→97.5, brain MRI 96.4→96.8), showing that refinement contributes on top of the prompt itself.

#### 4.5.2 EFFECT OF MASKING RATIO

Figure 4 isolates the efficiency claim. MWM reaches 96.8% accuracy on chest CT while masking only 40% of the image—well below the roughly 70% standard in masked image modeling—and still beats SparK’s best configuration by +2.4 absolute points. The comparison is deliberately conservative: we sweep SparK across 40%, 50%, 60%, and 70% and find its accuracy peaks at 70%, so MWM’s lower-ratio result is measured against SparK’s *strongest*, not a handicapped, setting. Put plainly, MWM does more with less—higher accuracy at nearly half the masking. This overturns the common assumption that high masking ratios are inherently optimal: once masking is semantically guided, preserving task-relevant content and dropping redundant background delivers better downstream performance at a substantially lower ratio.

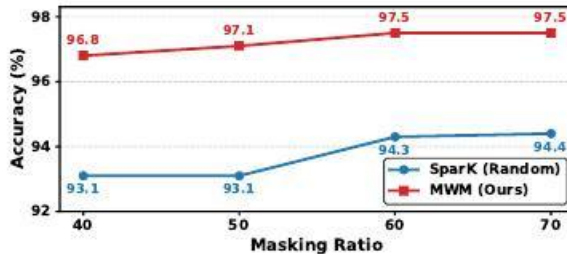


Figure 4: Chest CT classification accuracy versus masking ratio for MWM and SparK. MWM peaks at 96.8% with only 40% masking, exceeding SparK’s best result—obtained at 70%—by +2.4 points. SparK is swept over 40/50/60/70% and peaks at 70%, confirming the comparison uses SparK’s own optimal setting.

## 5 CONCLUSION AND LIMITATIONS

This work presents **Mask What Matters (MWM)**, a self-supervised pretraining framework that integrates text-guided semantic localization with region-aware masking. By using natural language prompts to highlight task-relevant areas and applying differentiated masking, MWM enables semantically aligned representation learning without per-image annotations.

The experiments support three quantitative conclusions, each verified across all three modalities under a controlled protocol: (1) text-guided localization reliably recovers task-relevant regions (around 0.82–0.94 occlusion recall), validating the feasibility of text-driven masking; (2) MWM outperforms the strongest MIM baselines (SparK, AnatoMask) on *every* modality and task tested—up to +3.1 points in fine-tuning classification and up to +6.9 in linear probing over SparK, plus +1.3 BoxAP and +1.1 MaskAP in detection/segmentation—so the advantage is a consistent pattern rather than an isolated result; (3) MWM attains these gains while masking only 40% of the image, beating SparK’s best 70% configuration by +2.4 points and thus delivering higher accuracy at markedly lower cost.

**Limitations.** These gains come with caveats that temper, but do not overturn, the quantitative picture. First, while prompt style clearly affects results—as the modality-dependent rankings in Tables 1 and 5 show—we have not systematically stress-tested robustness to prompt vocabulary or noise. Second, the reliance on external textual descriptions is a dependency that could weaken generalization where reliable prompts are unavailable or inconsistent. Importantly, even the weakest text-guided configuration in our ablations still exceeds random masking on every modality, suggesting the approach is not brittle to reasonable prompt choices.

Looking forward, text-driven, region-aware masking offers a principled path toward more semantically grounded self-supervised learning, and the uniform, efficiency-preserving gains reported here suggest that extending MWM beyond medical imaging is a promising direction for broader vision-language pretraining.

## REFERENCES

- Sana Aleem, Feiran Wang, Midhun Maniparambil, Eric Arazo, Johannes Dietlmeier, Kevin Curran, Krzysztof J. Geras, Sotirios A. Tsaftaris, Simon J. D. Prince, William M. Wells, William Lotter, Seung Wook Kim, Sebastian Pölsterl, and Samuel Little. Test-time adaptation with salip: A cascade of sam and clip for zero-shot medical image segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 5184–5193, 2024.
- Sourya Basu, Manan Gupta, Chahat Madan, Pulkit Gupta, and Chetan Arora. Focusmae: Gallbladder cancer detection from ultrasound videos with focused masked autoencoders. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 11715–11725, 2024.
- Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A simple framework for contrastive learning of visual representations. In *Proceedings of the 37th International Conference on Machine Learning (ICML)*, pp. 1597–1607. PMLR, 2020a. URL <https://proceedings.mlr.press/v119/chen20j.html>
- Xinlei Chen, Haoqi Fan, Ross Girshick, and Kaiming He. Improved baselines with momentum contrastive learning. *arXiv preprint arXiv:2003.04297*, 2020b.
- Jin Cheng, Wei Huang, Shuangliang Cao, Riqiang Yang, Wenjia Yang, Zhaoqiang Yun, and et al. Enhanced performance of brain tumor classification via tumor region augmentation and partition. *PLOS ONE*, 10(10):e0140381, 2015.
- Md. E. H. Chowdhury, Tanzila Rahman, Ahsan Khandakar, Rashid Mazhar, Md. Abdul Kadir, Zaid B. Mahbub, and et al. Can AI help in screening viral and COVID-19 pneumonia? *IEEE Access*, 8:132665–132676, 2020.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, volume 1, pp. 4171–4186, 2019.
- Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020.
- Ali Fateh, Yalda Rezvani, Sadegh Moayedi, Sam Rezvani, Farzaneh Fateh, and Mohammad Fateh. Brisc: Annotated dataset for brain tumor segmentation and classification with swin-hafnet. *arXiv preprint arXiv:2506.14318*, 2025.
- Fernando Feltrin. Brain tumor mri images 44 classes, 2023.
- Jean-Bastien Grill, Florian Strub, Florent Alché, Corentin Tallec, Pierre H. Richemond, Elena Buchatskaya, and et al. Bootstrap your own latent: A new approach to self-supervised learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pp. 21271–21284, 2020.

- Amit Gupta, Islam Osman, Mohamed S. Shehata, William J. Braun, and Richard E. Feldman. Medmae: A self-supervised backbone for medical imaging tasks. *Computation*, 13(4):88, 2025.
- Ahmed Hamada. Br35h :: Brain tumor detection 2020, 2025.
- Mohamed Hany. Chest ct-scan images dataset for lung cancer classification, 2020.
- Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 770–778, 2016.
- Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 16000–16009, 2022.
- Stanley C. Huang, Abhishek Pareek, Michael Jensen, Matthew P. Lungren, Serena Yeung, and Akshay S. Chaudhari. Self-supervised learning for medical image classification: A systematic review and implementation guidelines. *NPJ Digital Medicine*, 6(1):74, 2023.
- Anukriti Jaiswal, Abhinav Rajendra Babu, Mojtaba Zadeh Zadeh, Debanjan Banerjee, and Fillia Makedon. A survey on contrastive self-supervised learning. *Technologies*, 9(1):2, 2020.
- Daniel Kermay, Kang Zhang, and Michael Goldbaum. Labeled optical coherence tomography (oct) and chest x-ray images for classification. Mendeley Data, v2, 2018. URL <https://doi.org/10.17632/rschjbr9sj.2>.
- Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, and et al. Segment anything. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 4015–4026, 2023.
- Tareq Koleilat, Hamed Asgariandehkordi, Hassan Rivaz, and Yuyin Xiao. Medclip-sam: Bridging text and image towards universal medical image segmentation. In *Proceedings of the International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pp. 643–653, Cham, 2024. Springer Nature Switzerland.
- Sandor Konya. Ct lung & heart & trachea segmentation, 2020.
- Rahul Krishnan, Pranav Rajpurkar, and Eric J. Topol. Self-supervised learning in medicine and healthcare. *Nature Biomedical Engineering*, 6(12):1346–1352, 2022.
- Yufei Li, Tianyi Luan, Yuhang Wu, Shuhao Pan, Ying Chen, and Xujie Yang. Anatomask: Enhancing medical image segmentation with reconstruction-guided self-masking. In *Proceedings of the European Conference on Computer Vision (ECCV)*, pp. 146–163, Cham, 2024. Springer Nature Switzerland.
- Xin Liu, Fan Zhang, Zhenyong Hou, Lingqiao Mian, Zi Huang, Jian Zhang, and Jinhui Tang. Self-supervised learning: Generative or contrastive. *IEEE Transactions on Knowledge and Data Engineering (TKDE)*, 35(1):857–876, 2021.
- Stuart Lloyd. Least squares quantization in pcm. *IEEE Transactions on Information Theory*, 28(2): 129–137, 1982.
- Bjoern H. Menze, Andras Jakab, Stefan Bauer, Jayashree Kalpathy-Cramer, Keyvan Farahani, Justin Kirby, and et al. The multimodal brain tumor image segmentation benchmark (brats). *IEEE Transactions on Medical Imaging*, 34(10):1993–2024, 2014.
- Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, pp. 8748–8763. PMLR, 2021.

- Tanzila Rahman, Ahsan Khandakar, Yazan Qiblawey, Abdulla Tahir, Serkan Kiranyaz, Sheikh Abul Kashem, Md. Islam, Somaya Al Maadeed, Samar Zughaier, M. A. Khan, and Md. E. H. Chowdhury. Exploring the effect of image enhancement techniques on COVID-19 detection using chest x-rays images. *Computers in Biology and Medicine*, pp. 104319, 2021. doi: 10.1016/j.combiomed.2021.104319.
- Kai Tian, Yifan Jiang, Qihang Diao, Chaojian Lin, Li Wang, and Zehuan Yuan. Designing BERT for convolutional networks: Sparse and hierarchical masked modeling. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2023.
- Haoran Wang, Kaiyang Song, Jiashuo Fan, Yibing Wang, Jingdong Xie, and Zhenzhong Zhang. Hard patches mining for masked image modeling. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 10375–10385, 2023a.
- Ying Wang, Tim G. J. Rudner, and Andrew Gordon Wilson. Visual explanations of image-text representations via multi-modal information bottleneck attribution. In *Thirty-seventh Conference on Neural Information Processing Systems*, 2023b. URL <https://openreview.net/forum?id=ECvtxmVP0x>
- Jing Xiao, Yufan Bai, Alan Yuille, and Zongwei Zhou. Delving into masked autoencoders for multi-label thorax disease classification. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pp. 3588–3600, 2023.
- Shuai Zhang, Yiqiu Xu, Naoya Usuyama, Hongfang Xu, Jayesh Bagga, Rema Tinn, et al. Biomed-clip: A multimodal biomedical foundation model pretrained from fifteen million scientific image-text pairs. *arXiv preprint arXiv:2303.00915*, 2023.
- Yiwu Zhong, Jianwei Yang, Pengchuan Zhang, Chunyuan Li, Noel Codella, Liunian Harold Li, Luowei Zhou, Xiyang Dai, Lu Yuan, Yin Li, and Jianfeng Gao. Regionclip: Region-based language-image pretraining. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 16772–16782, 2022. doi: 10.1109/CVPR52688.2022.01629.

## A USE OF LARGE LANGUAGE MODELS (LLMs)

We used LLMs *only* for language editing and presentation polishing (e.g., grammar, phrasing, and minor stylistic clarity) of text written by the authors. LLMs were *not* used for idea generation, experimental design, data annotation, code implementation, or analysis/interpretation of results. All technical content, algorithms, proofs, figures, tables, metrics, and conclusions were authored and verified by the authors. No confidential or identifying data were provided to LLMs beyond anonymized manuscript text.